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PROJECT 1 PROPOSAL

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1.0 PROJECT TITLE

Our proposed title for this project is **Business Intelligence Solution for Retail Pricing and Demand Analysis Using Alteryx and Power Tableau**.

2.0 PROBLEM BACKGROUND

In today's competitive retail industry, businesses fruit a large amount of data related to product pricing, customer demand and sales transactions. However, raw data alone is not sufficient to support active decision-making unless it is properly integrated, processed and analyzed. Retail companies need a business intelligence tool and solution that can transform complex datasets into meaningful insights to improve business performance and operational efficiency.

This project aims to develop a BI solution by integrating multiple retail-related datasets, including product catalog data, pricing and demand signals, and sales records. By utilizing Alteryx Designer for data preparation and Power BI/Tableau for visualization, the project seeks to analyze sales trends, pricing behavior, and product demand patterns to support data-driven decision-making in the retail sector.

2.1 Business Context

The retail industry operates in a highly competitive and rapidly changing environment where customer demand, pricing strategies, and product performance constantly fluctuate. Retail companies must make fast and accurate business decisions to remain competitive, maximize profits, and satisfy customer needs.

Retailers often collect large amounts of data from different sources such as product catalogs, sales transactions, and pricing records. However, these datasets are usually separated and difficult to analyze effectively without a proper Business Intelligence (BI) solution.

This project focuses on integrating retail product catalog data, pricing and demand signal data, and sales transaction data to provide meaningful insights into product performance, customer demand patterns, and pricing effectiveness. By using BI tools such as Alteryx Designer and Power BI/Tableau, organizations can transform raw retail data into actionable business intelligence for strategic decision-making.

2.2 Importance Of The Problem

One of the major challenges faced by retailers is understanding how pricing and demand influence sales performance. Incorrect pricing strategies may lead to reduced profits, low customer satisfaction, or unsold inventory. At the same time, poor demand analysis may cause overstocking or stock shortages, affecting operational efficiency and customer experience.

In many organizations, data is stored across multiple systems and formats, making it difficult to generate accurate and timely insights. Without proper data integration and analysis, businesses may struggle to identify high-performing products, monitor sales trends, and optimize pricing strategies.

Therefore, implementing a BI solution is important because it enables retailers to:

- Analyze sales and demand trends more efficiently
- Improve pricing decision-making
- Identify top-performing and underperforming products
- Support strategic planning using data-driven insights
- Enhance operational and business performance

2.3 Expected Impact

This project is expected to help retail businesses gain a better understanding of product demand, pricing behavior, and sales performance through interactive dashboards and analytical reports.

The proposed BI solution will provide several positive impacts, including:

- Improved decision-making through data visualization and analytics
- Better pricing optimization strategies to maximize revenue
- Enhanced understanding of customer demand patterns
- Faster identification of sales trends and business opportunities
- More efficient data management through integrated datasets

Additionally, the project demonstrates how Business Intelligence technologies can assist organizations in transforming raw retail data into valuable insights that support business growth and competitiveness.

3.0 OBJECTIVES

1. To design and implement an end-to-end Business Intelligence solution that combines retails pricing signals, product catalog metadata, and historical sales performance.
2. To develop a workflow that prepares a dataset for visualization by conducting data cleansing & transformation tasks.
3. To identify seasonal peaks and low performing product categories by analyzing sales trends across 2023-2024 and their demand signals across different product segments.
4. To develop an interactive visualization that allows stakeholders to see which products are most sensitive to price changes, and which product is a slow-moving stock.
5. To provide insights for future enhancement sales of the retail store shopping.

4.0 DATASET DESCRIPTION

The dataset description explained in detail about the dataset that was used in this project. It explains about the dataset sources, type of data, shape of the data and the data column description where listed out all the column names, data type and description of the dataset.

4.1 Retail Pricing & Demand Signals dataset

- Source : Kaggle - Retail Pricing & Demand Signals Dataset
- Type of data: Structured
- Shape: 172,780 rows x 17 columns

Data column description;

Column name	Data type	Description
date	Object (String)	Date of the pricing and demand observation
product_id	Object (String)	Unique identifier for the product
category	Object (String)	Product category
brand	Object (String)	Product brand
region	Object (String)	Geographic market or region

channel	Object (String)	Sales channel such as web, mobile, or app
season	Object (String)	Seasonal period associated with the date
base_price	Float64 (Decimal)	Original product price before promotion or discount
current_price	Float64 (Decimal)	Effective selling price after discount
price_change_pct	Float64 (Decimal)	Percentage change from base price
discount_pct	Int64 (Integer)	Discount percentage applied to the product
promotion_type	Object (String)	Type of promotion applied
units_sold	Int64 (Integer)	Number of units sold during the observation period
revenue	Float64 (Decimal)	Total revenue generated from units sold
inventory_level	Int64 (Integer)	Available inventory after sales activity
stockout_flag	Boolean	Indicates whether inventory is critically low

demand_index	Float64 (Decimal)	Normalized demand score for comparison across products

4.2 Retail Product Catalogue dataset

- Source : Kaggle - Retails Product Catalogue
- Type of data: Structured
- Shape: 20 rows × 11 columns

Data column description;

Column name	Data type	Description
product_id	Int64 (Integer)	Unique numeric identifier assigned to each product in the catalog
product_name	Object (String)	Full descriptive name of the product as listed in the retail catalog
brand	Object (String)	Manufacturer or brand name associated with the product

category	Object (String)	Top-level product category (e.g., Groceries, Electronics, Clothing, Home & Kitchen, Personal Care, Sports & Outdoors)
sub_category	Object (String)	More specific classification within the main category (e.g., Dairy, Audio, Skincare, Fitness, Cookware)
price	Float64 (Decimal)	Listed retail price of the product in USD (ranges from $3.49 to $599.99)
rating	Float64 (Decimal)	Average customer rating of the product on a scale of 1.0 to 5.0
review_count	Int64 (Integer)	Total number of customer reviews submitted for the product
stock_status	Object (String)	Current inventory availability status of the product (e.g., In Stock, Out of Stock)
tags	Object (String)	Pipe-separated keywords associated with the product used for search and recommendation purposes
description	Object (String)	Short textual description summarising the product features and specifications

4.3 Product Sales dataset

- Source : Kaggle - [Product Sales]
- Type of data: Structured
- Shape: 200,000 rows × 14 columns

Data column description;

Column name	Data type	Description
Order_ID	Int64 (Integer)	Unique numeric identifier assigned to each sales order transaction
Order_Date	Object (String)	Date on which the order was placed, covering the period from January 2023 to December 2024
Customer_Name	Object (String)	Full name of the customer who placed the order
City	Object (String)	City in which the customer or delivery address is located
State	Object (String)	US state corresponding to the customer or delivery address

Region	Object (String)	Geographic US region of the order (South, Centre, East, West)
Country	Object (String)	Country of the order destination (United States)
Category	Object (String)	Top-level product category of the sold item (Accessories, Clothing & Apparel, Electronics, Home & Furniture)
Sub_Category	Object (String)	Specific sub-category within the broader product category (e.g., Smartphones, Sportswear, Furniture, Kitchenware)
Product_Name	Object (String)	Name of the specific product associated with the order line
Quantity	Int64 (Integer)	Number of units ordered in the transaction (ranges from 1 to 11)
Unit_Price	Float64 (Decimal)	Price per unit of the product at the time of sale (ranges from $17.03 to $1,432.00)
Revenue	Float64 (Decimal)	Total revenue generated from the order line (Quantity x Unit_Price), ranging from $17.03 to $9,014.25

Profit	Float64 (Decimal)	Net profit earned from the order line after costs, ranging from $3.92 to $2,763.72

5.0 PROPOSED BUSINESS INTELLIGENCE APPROACH

5.1 Use of Alteryx and Visualization Tools

Alteryx Designer will serve as the primary data preparation platform for this particular project, involving the full pipeline from loading and cleaning the datasets through to transformation and integration. Then, Tableau will be used to build the interactive dashboards and storyboard that present the final analysis to stakeholders.

Alteryx is chosen for its visual, drag-and-drop workflow interface, which then makes it straightforward to design, and adjust a multi-step data preparation process without writing the codes. This is important for a project that works across three datasets with different granularities and structures. Workflows in Alteryx canvas make the logic of every transformation visible at a glance, and connecting all data sources through Join and Union tools is a well-supported process. Each tool in the workflow performs one specific job, which also makes it easier to spot and fix errors if the data produces unexpected results at any stage.

Tableau is selected as the visualization layer as it supports direct connection to CSV outputs from Alteryx and produces interactive dashboards where users can apply filters, switch views, and dig into specific categories. The project will produce at least two dashboards and one storyboard in Tableau, covering pricing and performance, product-level insights from the catalog, and regional sales trends, satisfying Phase 3 deliverable requirements. Both tools are widely used in industry BI workflows, and their combination represents a complete extract-transform-load-visualize pipeline that provides real-world retail analytics practice.

5.2 Data Integration, Cleaning, and Transformation

5.2.1 Data Integration

All three datasets will be loaded into Alteryx separately and prepared on their own branches before being merged together. This action will keep the pipeline clean so that any issues in one dataset can be fixed without affecting the others.

Dataset 1 (Retail Pricing & Demand Signals) will act as the main table. It will hold the most granular data which are daily product-level pricing, discount, inventory, and demand observations, so it makes sense to use it as the base that the other two datasets rely on.

Dataset 2 (Retail Product Catalog) will be joined to Dataset 1 using product_id as the join key. Since both are published by the same author on Kaggle, the product identifiers are consistent across both files, which makes the join straightforward. This adds product name, brand, and category metadata to each pricing observation, which is the context that Dataset 1 alone does not carry together.

Dataset 3 (Product Sales 2023-2024) covers the transaction-level order records including region, category, and revenue. Since it comes from a different source, the plan is to join it to the merged table at the category level rather than at the product level., using aggregated monthly totals per category as the connecting dimension.

This is the most meaningful way to link both without forcing an artificial row-level match between datasets that were not designed to share a common key.

5.2.2 Data Cleaning

Before the involved datasets are merged, they will go through a cleaning pass to ensure the data that are going into the integration step is cleaned, consistent, and reliable. The planned cleaning steps are:

- Handling missing values: Records with nulls in keyfields such as product_id, category, date, and revenue will be removed. For less critical fields like promotion_type, missing values will then be replaced with a default label rather than dropping the whole row.
- Removing duplicates: Each dataset will be checked for duplicate rows and make sure they are deduplicated before merging. For Dataset 1, duplicates will be identified based on the combination of date, product_id, and region.
- Filtering invalid records: Rows with logically impossible values like negative revenue, zero units sold, and prices that do not make sense will be dropped out as likely data entry errors.
- Fixing data types: The date columns in Datasets 1 and 3 are stored as strings in their raw files. These will have to be converted to proper DateTime format so that the time-based grouping and Tableau's date axis work correctly.

5.2.3 Data Transformation

After cleaning, few new columns will then be derived from the existing fields to make the data more insightful for the analysis. The planned transformations are:

- month and year: Extracted from the date fields in Datasets 1 and 3 to support monthly trend analysis across 2023–2024.
- discount_amount: The difference between base_price and current_price in Dataset 1, giving a monetary figure for the discount applied rather than just a percentage.
- price_sensitivity_flag: A label derived from price_change_pct and units_sold in Dataset 1 that marks products as "High Sensitivity" where price increases correspond with notable drops in sales volume.
- stock_status: A readable category derived from inventory_level and stockout_flag — grouping products as "In Stock", "Low Stock", or "Out of Stock" — so inventory risk is visible without needing to interpret raw numbers.
- revenue_per_unit: Calculated from Dataset 3 to show the average selling price per item by category, independent of how many units were sold.

- product_tier: Products in Dataset 2 will be grouped into tiers (e.g. Premium, Mid-Range, Budget) based on catalog price ranges, adding a useful segmentation dimension for comparison across dashboards.

Once all of the transformations are done, the data will then be aggregated by category, season, region, and month, then exported as a CSV for Tableau to pick up.

5.3 Analysis Plan and Proposed Visual Forms

5.3.1 Line Chart: Monthly Revenue and Sales Trend (2023-2024)

A line chart will be used to show how units sold and total revenue changed monthly across two years. The x-axis will run through each month from 2023 to 2024, and the y-axis will show revenue, with a second axis overlaid for units sold so both trends can be read in the same view. The reason for choosing a line chart is, it is the most natural way to show something moving through time, and any seasonal peaks will be visible in the shape of the line. If revenue rises while units sold stay flat, that would also show up as a divergence between the two lines, which would be worth investigating.

5.3.2 Grouped Bar Chart: Revenue and Units Sold by Product Category and Tier

A grouped bar chart will compare total revenue and total units sold across all product categories, with bars colour-coded by product tier (Premium, Mid-Range, Budget) from Dataset 2. Bars will be sorted in descending order by revenue so the strongest categories stand out first. The grouped format is chosen over a stacked bar because the point is to compare categories directly against each other, not to show proportions within a whole. Adding the product tier dimension also makes the chart more informative like it would show whether a high-revenue category is being driven by a small number of premium products or by large volumes of budget items.

5.3.3 Scatter Plot: Price Change vs. Units Sold

A scatter plot will explore the relationship between price_change_pct and units_sold using Dataset 1. Each point on the chart represents one product observation and the points will then be coloured by category so that pattern with individual segments will be visible. Products that react instantly to price increases mean that their sales volume drops a lot when prices go up and it will cluster toward the lower-right area of the chart and stand out as high-sensitivity items. A trend line will be added to mirror the overall direction of the relationship. This is the right chart for this type of analysis as scatter plots are built for spotting correlations and distributions between two numerical variables, which no bar or line chart can replicate.

5.3.4 Heatmap: Seasonal Demand by Product Category

A heatmap will display total units sold in a grid where each row is a product category and each column is a season. Darker cells mean higher demand for that specific category-season combination. The reason for using a heatmap rather than multiple bar charts is efficiency as comparing demands across all four seasons for each category at one will need a lot of separate charts, but a heatmap condenses it all into one view that is easy to scan. High-demand and low-demand combinations jump out immediately through the colour contrast, which then directly supports the project's goal of identifying seasonal peaks and underperforming periods.

5.3.5 Treemap: Inventory Distribution and Slow-Moving Stock

A treemap will show the inventory distribution of all products, where each rectangle's size reflects the product's current inventory level. Products flagged as "Low Stock" or "Out of Stock" from the stock_status derived column will be highlighted in a different colour so they are easy to spot. Product names and brands from Dataset 2 will be used to label each rectangle, making the chart something stakeholders can act on rather than just look at. A bar chart would get too crowded at this scale, but a treemap will handle a large number of products cleanly because the proportional layout does the comparison work visually. This is the most practical way to give a quick overview of where inventory risk sits across the full product range.

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